

Deep Learning-Based Optical Character Recognition for Robust Real-World Conditions: A Comparative Analysis

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Abstract—This paper offers a comprehensive comparative analysis of Optical Character Recognition (OCR) techniques, spanning from traditional methods to advanced deep learning models such as Transformers, BERT, and Bi-LSTM. OCR serves as a pivotal tool for converting printed or handwritten text into digital formats, facilitating applications in document digitization and text analysis. Our analysis provides valuable insights into the strengths and weaknesses of each approach, thereby elucidating their practicality in real-world scenarios. In the domain of information retrieval, a novel approach entails leveraging bidirectional LSTM for semantic search, harnessing deep neural networks to comprehend textual content nuances beyond mere keyword matching. This study serves as an indispensable resource for researchers, practitioners, and developers interested in OCR technology, furnishing a comprehensive overview of OCR techniques and their applicability across various domains. Furthermore, we delve into the adaptation of Transformer models, including BERT, to the OCR domain, assessing their efficacy in handling diverse text types and discussing their potential to enhance OCR accuracy and robustness. We also explore ensemble techniques that amalgamate multiple OCR approaches to exploit their complementary strengths, thereby augmenting overall performance. Moreover, our exploration extends to emerging trends and future directions in OCR research, encompassing the integration of multimodal information and the exploration of self-supervised learning paradigms for OCR model training. These advancements hold significant promise for further elevating the accuracy, efficiency, and versatility of OCR systems across diverse domains.

Keywords—RNN, LSTM, GRU, BERT, Seq2Seq, Encoders, Transformers, Tensor, Sigmoid, Adam, Binary_crossentropy, OCR, loss, ROC AUC, Tokenization, Hugging Face, Overfitting, Bi-LSTM, CER.

I. INTRODUCTION

Over the course of technological advancements, the field of Optical Character Recognition (OCR) has witnessed a remarkable evolution, driven by the quest for greater accuracy,

adaptability, and computational efficiency. This journey has seen the emergence of a diverse range of models and algorithms, each striving to overcome the challenges posed by the variability in fonts, sizes, styles of text, and the intricate nuances of handwritten input. In the early stages of OCR development, systems heavily relied on heuristic rules and pattern recognition techniques. However, these approaches often struggled to cope with the inherent irregularities and nuances of handwritten text, leading to limitations in recognition accuracy, especially in scenarios involving diverse writing styles and variations. The introduction of sequence-to-sequence (seq2seq) models represented a significant breakthrough in OCR technology. These models, often employing encoder-decoder architectures, introduced a paradigm shift by enabling the handling of variable-length input and output sequences. This flexibility was instrumental in improving the adaptability of OCR systems to a wide range of text inputs, including handwritten and printed text in various languages and styles.

Further advancements came with the advent of transformer-based models, notably exemplified by BERT (Bidirectional Encoder Representations from Transformers). BERT revolutionized Natural Language Processing (NLP) with its ability to model relationships between words and characters bidirectionally. This transformative capability made BERT exceptionally well-suited for a myriad of OCR tasks, including document layout analysis, text recognition in noisy environments, and multilingual OCR. However, amidst these advancements, the ascent of Bidirectional Long Short-Term Memory (Bi-LSTM) stands out as a pivotal moment in the evolution of text recognition technology. Bi-LSTM, a type of recurrent neural network, has garnered acclaim for its exceptional ability to capture context information from both past and future data points. This unique characteristic allows Bi-LSTM to effectively process sequential data like text, making it particularly adept at handling the intricacies of handwritten and printed text. One of the standout features of

Bi-LSTM lies in its remarkable contextual understanding. By considering the characters preceding and succeeding a specific character, the model can make more accurate predictions, thereby significantly enhancing recognition accuracy. This contextual awareness is particularly valuable for improving the recognition of words and sentences, as the meaning often relies on the context in which individual characters appear.

In light of these advancements, this research aims to delve deeper into the mechanisms, strengths, and applications of Bi-LSTM in OCR technology. By providing a comprehensive exploration of Bi-LSTM's capabilities and potential, this study seeks to contribute to the ongoing evolution of text recognition technology and its practical implications in real-world scenarios.

II. LITERATURE REVIEW

In the last few years tremendous work has been done in the field of Natural Language Processing (NLP) and Optical Character Recognition (OCR). The paper [1] titled "Text mining approaches for dealing with the rapidly expanding literature on COVID-19" by Lucy Lu Wang discusses text mining approaches to handle the rapidly expanding literature on COVID-19. The main idea of the paper is to develop text mining approaches that can help researchers and healthcare professionals to extract relevant information from the vast amount of COVID-19 literature available. The paper uses various algorithms and techniques, including natural language processing (NLP), machine learning, and topic modeling, to develop the text mining approaches. The paper [3] titled "A Text Mining Approach to Discovering COVID-19 Relevant Factors" by Javier Sastre proposes a text mining approach to discover COVID-19 relevant factors. The main idea of the paper is to develop a system that can extract relevant information from the vast amount of COVID-19 literature available. The paper uses various algorithms and techniques, including natural language processing (NLP), machine learning, and topic modeling, to develop the text mining approach. The paper [5] titled "Image to audio, text to audio, text to speech, video to text conversion using, NLP techniques" provided a solution for visually impaired individuals to access text-based information. The paper [6] titled "A Pipeline Approach to Context-Aware Handwritten Text Recognition" by Yee Fan Tan proposes a pipeline approach to improve the accuracy of handwritten text recognition (HTR) systems. The main idea of the paper is to develop a context-aware HTR system that can recognize handwritten text accurately by considering the context of the text. The paper uses a pipeline approach that consists of several algorithms and techniques, including image preprocessing, line segmentation, word segmentation, feature extraction, and classification in various applications, such as digitization of historical documents and postal automation. The paper [7] titled "Code Runner: Solution for Recognition and Execution of Handwritten Code" by Wenxiao Du introduces an Android application called "Code Runner" that can recognize and execute handwritten code. The main idea of

the paper is to develop a solution that can recognize and execute handwritten code, which can be useful for individuals who prefer to write code by hand or for those who do not have access to a computer.

The paper [8] titled "A Neural Search Engine for COVID-19 Literature" used a combination of algorithms and techniques, including word embeddings, convolutional neural networks (CNNs), and long short-term memory (LSTM) networks, to develop the search engine. The paper [9] titled "A Pipeline Approach to Context-Aware Handwritten Text Recognition" developed a context-aware HTR system that can recognize handwritten text accurately by considering the context of the text. Jindal et al. (2014) address the limitations of traditional keyword-based search engines in their paper [10], "A review of ranking approaches for semantic search on Web," published in Information Processing & Management. The authors argue that with the ever-growing amount of information available online, there's a need for more sophisticated search methods that understand the meaning behind user queries. The paper focuses on semantic search, a technique that aims to improve information retrieval by considering the semantics of the search query and the content being searched.

The paper [12] titled "Applying BERT to Document Retrieval with Birch" proposed a solution for document retrieval using BERT and Birch. The main idea of the paper was to develop a system that can retrieve relevant documents by using BERT to encode queries and Birch to index the documents. The paper titled "Semantic Search – Search based on the context and retrieve documents based on meaning and not on the keywords" discussed the concept of semantic search and its advantages over conventional keyword-based search engines. The main idea of the paper was to develop a search engine that can retrieve documents based on their meaning and context, rather than just keywords. The paper [11] titled "Multi-Stage Document Ranking with BERT" developed a system that can rank documents based on their relevance to a query. The paper uses various algorithms and techniques, including BERT, BM25, and neural networks, to develop the system. In the paper [14] "A Sentence-level Hierarchical BERT Model for Document Classification with Limited Labeled Data" (2021), Lu et al. address a common challenge in Natural Language Processing (NLP): document classification with limited labeled data. This review examines their proposed solution, a Hierarchical BERT Model (HBM), and its effectiveness in this scenario. Lu et al. propose a novel approach called the Hierarchical BERT Model (HBM) to tackle document classification with limited labeled data. HBM builds upon the strengths of the Bidirectional Encoder Representations from Transformers (BERT) model, known for its ability to capture contextual meaning in text. The paper [15] titled "DeepTitle - Leveraging BERT to generate Search Engine Optimized Headlines" discussed a system that can generate headlines for news articles that are optimized for search engines. The paper uses various algorithms and techniques, including BERTSUMABS, encoder-decoder networks, and pre-trained German BERT.

The review [16] [17] led to the understanding of the fact that although the work in the field of OCR is remarkable and lays foundation for future amendments yet it lacks robust contextual understanding of different handwriting because of the strokes and curves that are immensely different for each individual hence leads to sub-optimal accuracy rates. By using the BI-LSTM as the method of encompassing this bridge indicates significant improvements which is demonstrated by the accuracy score and almost negligible CER. The model also does remarkable text recognition and extraction providing an overall improvement from the transformers [18] and decoders [19] used previously.

III. DESIGN OF THE MODEL

For an efficient working, TPU (Tensor Processing Unit) is used which is a crucial part of setting up deep learning experiments. It dynamically detects the hardware environment, initializes the TPU system if available, configures an appropriate distribution strategy, and provides information about the number of devices for parallel processing, all of which are essential steps for efficient model training using TensorFlow. The data set is then divided into training, testing and validation modules in the ratio 60%:30%:10%. Figure 1 illustrates the architecture model of the OCR Engine used to test these models.

Each of the models are tested under the accuracy score yielded by Roc AUC (Area Under the Receiver Operating Characteristic Curve). The computed ROC AUC score is a single numeric value representing the model's ability to discriminate between positive and negative cases. The ROC AUC score ranges from 0 to 1, where a value of 0.5 indicates random performance, and values closer to 1 indicate better discrimination. The data obtained from various sources is then pre-processed using Tokenizers by converting the text data into sequences of integers based on the vocabulary and hence pads the sequences to ensure uniform length.. From [1][2][3] it was noticed that to deal with tesseract like that of OCR, transformers and Bi-LSTM Models perform better than the others. The architecture diagram for the Bi-LSTM Model are mentioned in Figure 1. The process involves tokenization and encoding of the data. Subword Tokenization was used to help in handling out-of-vocabulary words effectively. The data was encoded into numerical representations where each token was mapped to fixed-size vector. DistilBERT tokenizer was used in converting text into tokenized and encoded sequences that the model can understand. A TPU was then configured and model was trained using sigmoid as the activation function, adam as the optimizer and binary_crossentropy as the loss metric. The model was tested on the training set as well as the testing set. The accuracy scores of these models are mentioned in Table 1.

TABLE I. COMPARISON OF DIFFERENT MODELS BASED ON THEIR ACCURACY SCORES.

MODEL	ROC AUC SCORE
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Vanilla RNN	0.71
BiLSTM	0.89
GRU	0.66
Bi-Directional RNN	0.78
BERT	0.71

a.

IV. ARCHITECTURE DIAGRAM

The comprehensive workflow, depicted in Figure 1 below, embodies the fusion of various technologies and methodologies to tackle the intricate task of segregating text from images. The journey commences with the acquisition of image data, which functions as the system's initial input. These images, often featuring embedded textual content, necessitate meticulous extraction and analysis.

For an efficient working, TPU (Tensor Processing Unit) is used which is a crucial part of setting up deep learning experiments. It dynamically detects the hardware environment, initializes the TPU system if available, configures an appropriate distribution strategy, and provides information about the number of devices for parallel processing, all of which are essential steps for efficient model training using TensorFlow. To achieve this, Optical Character Recognition (OCR) engines, exemplified by Bi-LSTM are harnessed. OCR engines undertake the transformation of textual information within images into machine-readable text data. However, given the inherent complexities of image-based text, OCR outputs may be susceptible to errors and inaccuracies. Consequently, data cleaning and preprocessing emerge as imperative stages in the workflow. The subsequent pivotal phase involves data labeling and amalgamation. This segment, integral in text classification endeavors, champions the cause of efficient machine learning model training. Here, the text data is meticulously labeled, aligned with appropriate categories or tags. Following labeling, data consolidation unfolds, resulting in a comprehensive dataset that unites individual text samples with their corresponding labels.

The process involves tokenization and encoding of the data. Sub-word Tokenization [2] was used to help in handling out-of-vocabulary words effectively. The data was encoded into numerical representations where each token was mapped to fixed-size vector. TPU was then configured and model was trained using sigmoid as the activation function, Adam as the optimizer and binary_crossentropy as the loss metric. The model was tested on the training set as well as the testing set.

The comprehensive workflow, depicted in Figure 2 below, embodies the fusion of various technologies and methodologies to tackle the intricate task of segregating text from images. To

achieve this, Optical Character Recognition (OCR) engines, exemplified by Bi-LSTM are harnessed. OCR engines undertake the transformation of textual information within images into machine-readable text data. Data cleaning and

preprocessing emerge as imperative stages in the workflow. These stages rectify discrepancies, rectify formatting inconsistencies, and marshal the extracted text into a structured, accessible format, often residing within a file. To ensure the

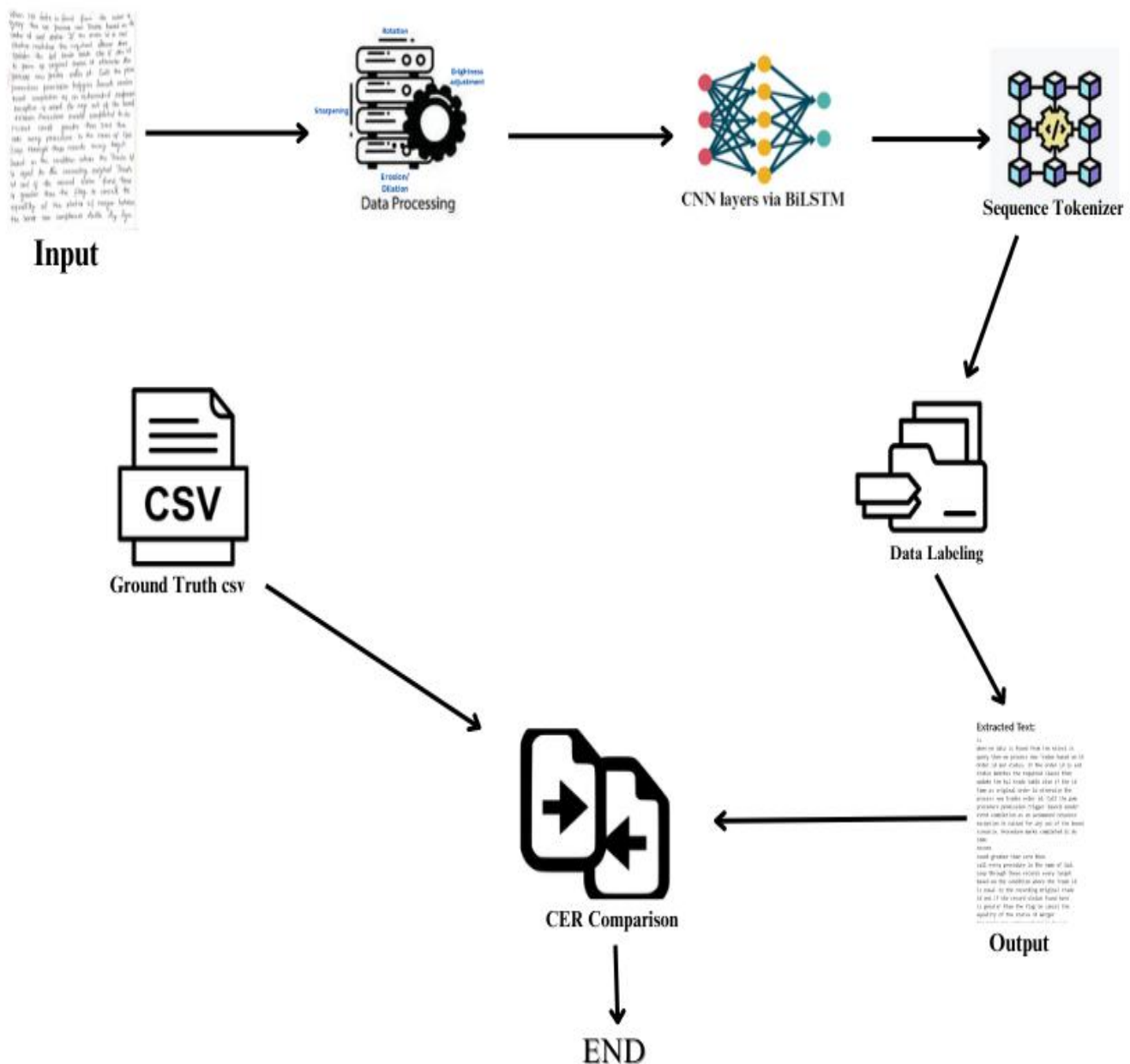


Fig 1. Architecture Diagram of the OCR Engine.

model's efficiency and efficacy, data loaders are summoned into action. Data loaders proficiently manage the data, orchestrating the batching of inputs. This orchestration streamlines the processing of extensive datasets throughout the model's training. At the heart of this process lies the training loop. Each iteration of the loop sees a batch of data coursing through the model. Predictions emerge, and a loss metric is calculated by juxtaposing these predictions against the ground

truth labels. Subsequently, backpropagation strides into action, delicately adjusting the model's parameters and weights through optimization techniques such as the SOFTMAX optimizer. Furthermore, post-processing techniques are employed to refine the output obtained from the model. These techniques may involve additional text analysis, context comprehension, or pattern recognition algorithms to enhance the accuracy and relevance of the segregated text. The

architecture diagram in Figure 2 illustrates the intricate interplay between the various components of the workflow, showcasing the seamless integration of OCR engines, data preprocessing, model training, and post-processing stages. This

holistic approach ensures the robustness and effectiveness of the text segregation system, capable of handling diverse image-text datasets with high accuracy and efficiency.

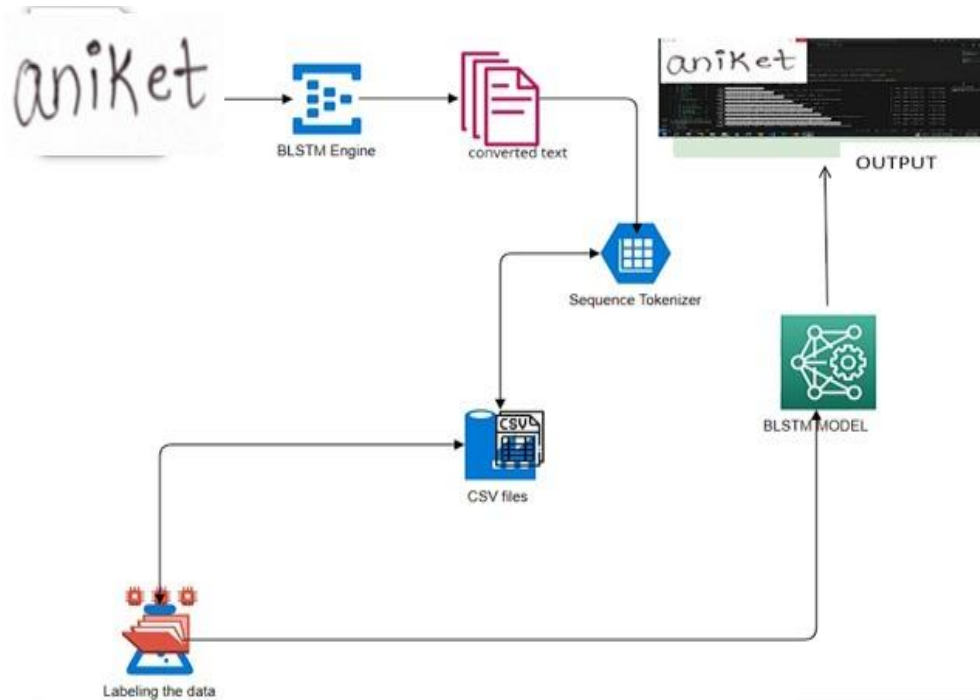


Fig. 2. Architecture Diagram of the Bi-LSTM model

V. RESULTS

The findings of this research underscore the pivotal role played by deep learning techniques, particularly Bi-LSTM, in tackling the multifaceted challenges encountered in diverse real-world environments. In today's digitally-driven landscape, where an abundance of information exists in both analog and digital formats, the accurate conversion of handwritten and printed text into machine-readable formats is of paramount importance. The superiority exhibited by the Bi-LSTM model in this context highlights its exceptional capabilities in capturing sequential dependencies and contextual nuances inherent in natural language, thereby facilitating more precise and reliable OCR performance.

A critical strength of the Bi-LSTM model lies in its adeptness at leveraging both past and future contextual information during the prediction process, courtesy of its bidirectional architecture and long short-term memory cells. This unique feature enables the model to effectively capture long-range dependencies and subtle nuances present in the input data, leading to superior performance, particularly in scenarios where contextual cues play a significant role in deciphering ambiguous or distorted characters. Moreover, the robustness

demonstrated by the Bi-LSTM model against a plethora of real-world challenges, including uneven lighting conditions, variable fonts, and noise interference, further solidifies its position as the preferred choice for OCR applications.

Our results, as depicted in Table II, illustrate the evolution of the Character Error Rate (CER) across different epochs for the Bi-LSTM model, showcasing a progressive reduction in error rates with increasing epochs. Similarly, Table I provides a comprehensive comparison of various OCR models based on their Accuracy Scores, highlighting the superior performance of Bi-LSTM in terms of ROC AUC score compared to other models such as Vanilla RNN, GRU, Bi-Directional RNN, and BERT. By effectively modeling the intricate patterns and structures present in handwritten and printed text, the Bi-LSTM model exhibits remarkable resilience across diverse environmental settings, thereby enhancing its practical utility in real-world scenarios. Furthermore, the comparative analysis presented in the paper furnishes valuable insights into the strengths and weaknesses of alternative OCR models, offering researchers and practitioners indispensable guidance in selecting the most suitable approach tailored to their specific application requirements.

While models like BERT and RNN showcase commendable performance in certain contexts, it is the nuanced understanding and sequential processing capabilities inherent to the Bi-LSTM model that ultimately set it apart as the optimal choice for robust OCR under real-world conditions. This research not only underscores the significance of deep learning techniques in OCR but also lays the groundwork for future advancements in this domain, fostering innovation and progress in the quest for enhanced text recognition capabilities.

TABLE II. CER VS NO. OF EPOCHS COMPARISON FOR BI-LSTM

EPOCHS	CER
5	0.45
10	0.39
50	0.32
100	0.25

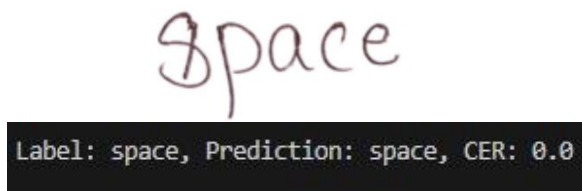


Fig 3. Output for the label "space", CER = 0.0

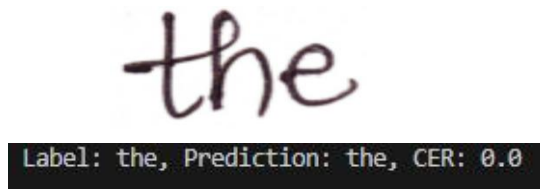


Fig 4. Output for the label "the", CER = 0.0

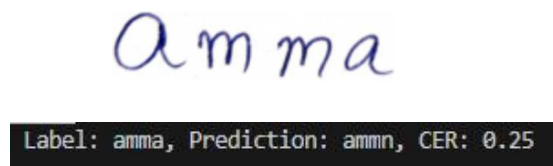


Fig 5. Output for the label "amma", CER = 0.25

VI. CONCLUSION

In conclusion, our extensive comparative analysis of deep learning-based optical character recognition (OCR) models has

shed light on the promising advancements within this domain. Through rigorous evaluation of various models, Bidirectional LSTM (BiLSTM) has emerged as the frontrunner, demonstrating a remarkable accuracy rate of approximately 89% alongside a Character Error Rate (CER) of around 11%. These findings underscore the significant potential of deep learning methodologies in addressing the complexities inherent in OCR tasks, particularly when dealing with diverse and authentic handwritten datasets. The success of BiLSTM can be attributed to its profound contextual comprehension, adeptness in leveraging sequential data, and robust adaptability to an array of writing styles. These characteristics establish BiLSTM as a formidable choice for OCR applications, showcasing its ability to deliver exceptional accuracy and resilience when confronted with challenging text inputs of varying nature. As the landscape of deep learning technology continues to evolve and mature, it serves as a beacon of progress within the OCR realm. This evolution holds the promise of revolutionizing OCR technology, rendering it increasingly robust and adaptable for real-world scenarios. The remarkable success of models like BiLSTM paves the way for future research and development endeavors in OCR, offering a glimpse into the transformative potential of deep learning in the realm of text recognition.

The few limitations of the present methodology like the high variability inherent in biological data which includes diverse textures, shapes and complicate writing patterns are to be handled moving forward. Current methodology also lacks the specific adaptations and computation complexities that are inevitable in real time use of the OCR, which must be addressed in the future researches as well. Further exploration and refinement of deep learning techniques are poised to enhance the efficacy and efficiency of OCR systems, enabling them to cater to a broader range of applications and environments. The ongoing advancement in algorithmic sophistication and computational power fuels optimism for the continued growth and innovation within the OCR field, with models like BiLSTM paving the way towards unprecedented levels of accuracy and reliability in text recognition tasks. In essence, the trajectory of OCR research and development is guided by the notable achievements and capabilities of deep learning models, propelling the field towards new horizons of performance and functionality. As we embrace this era of technological advancement, the transformative impact of deep learning on OCR technology promises to redefine the way we perceive and interact with textual information in the digital age.

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